

Breakout Swing Scanner

Daily Donchian Breakouts + Intraday ORB Breakouts
Pine Script v6 indicator designed for TradingView watchlists and Pine Screener

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Disclaimer: This document is for educational purposes only and is not financial advice. Markets involve risk. Always test ideas on paper and use position sizing and stop-loss rules appropriate to your risk tolerance.

What you get: a scanner-style indicator that outputs clear Long/Short signals and screener columns, plus optional chart visuals and alert conditions. Use it to shortlist breakout candidates from a watchlist (e.g., S&P; 500) and then confirm setups on the chart.

1. Executive Summary

The Breakout Swing Scanner is a Pine Script v6 **indicator** (not a backtest strategy) that detects breakout conditions and publishes them as simple 1/0 outputs (signals) suitable for scanning many symbols. It supports two styles:

- **Daily breakout swing:** closes above/below a prior N-bar Donchian high/low, optionally filtered by trend, volume, ADX, relative strength vs SPY, and market regime.
- **Intraday breakout swing:** triggers on an Opening Range Breakout (ORB) and/or a daily Donchian level intraday, optionally filtered by session VWAP, relative volume, daily trend context, ADX, relative strength vs SPY, and market regime.

The script is designed to work well as a **watchlist scanner** using TradingView's **Pine Screener** (the first 10 plots become screener columns) and as a **watchlist alert** (alert conditions can fire per symbol).

Best use: scanning a watchlist. You can also apply it to a single stock chart for confirmation, but its main value is quickly narrowing a large universe down to a manageable shortlist.

2. What the Script Does

At every completed bar, the script evaluates whether the current symbol meets breakout criteria. If it does, it outputs:

- **LongSignal** = 1 when a qualified bullish breakout occurs; otherwise 0.
- **ShortSignal** = 1 when a qualified bearish breakout occurs; otherwise 0.
- **ActiveLevelLong/Short** = the price level being used as the breakout threshold.
- Supporting metrics such as **VolRatio/RelVol**, **ADX**, **VWAPDev**, **RSRatio vs SPY**, and a **SuggestedStopPrice**.

Signals are computed on **bar close** (confirmed-bar logic) to avoid intrabar flicker. In intraday mode, the script can limit to **one signal per day per direction** to reduce repeated triggers.

3. Signal Logic

3.1 Mode selection (Auto / Daily / Intraday)

The script has a **Run mode** input. In **Auto**, it chooses:

- **Daily logic** on non-intraday timeframes (1D, 1W, 1M, etc.).
- **Intraday logic** on intraday timeframes (1m-240m).

You can override Auto with “Daily only” or “Intraday only” to force behavior.

3.2 Daily breakout swing (Donchian breakout)

Daily mode uses Donchian channels based on a lookback of N bars, excluding the current bar:

- Donchian High = highest high of the previous N bars.
- Donchian Low = lowest low of the previous N bars.

A bullish breakout triggers when price crosses above the Donchian High; bearish triggers when it crosses below the Donchian Low. You can choose whether breakout is based on **Close** (more conservative) or **High/Low** (more aggressive).

3.3 Intraday breakout swing (ORB / Daily Donchian / Both)

Intraday mode supports three breakout bases:

- **Opening Range:** the high/low of the first X minutes of the regular session.
- **Daily Donchian:** breaks the prior N-day Donchian level intraday.
- **Both (confirm):** requires OR to form and then breaks the stricter level (max/min of OR and daily level).

Intraday signals are typically used for earlier entries while still targeting multi-day follow-through. Daily context values (daily EMAs and daily ATR) are referenced from the **previous completed daily bar** to reduce repaint.

4. Filters and Confirmations

Filters are optional. Turning filters on reduces signal count but may improve quality.

Volume surge / Relative volume. Daily: Volume must be $\geq (\text{Volume SMA} * \text{multiplier})$. Intraday: $\text{RelVol} = \text{Volume} / \text{VolumeSMA}$ must be $\geq \text{multiplier}$. Higher multipliers produce fewer signals.

EMA trend filter. Daily: requires price above/below slow EMA and EMA-fast stacked above/below EMA-slow. Intraday: uses previous daily EMAs to confirm swing direction.

VWAP filter (intraday). Longs require price above session VWAP; shorts require price below. Helps avoid buying breakouts under value.

ADX filter (optional). Requires $\text{ADX} \geq \text{threshold}$. ADX is obtained via the Directional Movement Index (DMI) and measures trend strength. Higher thresholds reduce choppy signals but can delay early breakouts.

Relative strength vs SPY (optional). Prefers symbols outperforming SPY (ratio above its own moving average for longs; below for shorts).

Market regime vs SPY (optional). Longs only when SPY is above its slow EMA; shorts only when below. Helps align trades with broad tape.

Important: Filters interact. Enabling Trend + ADX + RS + Market can reduce signals significantly. If you get too few signals, loosen filters before changing the breakout definition.

5. Inputs Reference

All inputs are optional and have defaults. Adjust them to match your trading style and the timeframe you scan.

Group	Input	Default	Effect / Notes
Core	Run mode	Auto	Auto selects Daily on non-intraday timeframes and Intraday on intraday timeframes. Can be forced.
Breakout	Donchian lookback (donLen)	20	Higher = fewer, broader breakouts. Lower = more frequent signals.
Breakout	Fresh breakout only	true	Signals only on the crossing bar. Off = stays true while above/below level.
Breakout	Use Close for breakout	true	Close is stricter; High/Low is earlier but noisier.
Intraday	Intraday breakout mode	Opening Range	Choose ORB vs daily level vs Both (confirm).
Intraday	Opening range minutes	30	Larger window = fewer ORB signals, more structure.
Session	Only evaluate inside session	true	Restricts intraday logic to the session window (e.g., 09:30-16:00).
Session	Session window	0930-1600	Defines ORB/VWAP reset and in-session evaluation.
Filters	Volume filter	true	Requires volume/relvol expansion.
Filters	Volume SMA length	20	Longer = smoother baseline; shorter = more reactive.
Filters	Volume multiplier	1.5	Higher = stricter volume requirement.
Filters	EMA trend filter	true	Requires directional alignment with EMA fast/slow stack.
Filters	EMA fast	50	Faster trend measure.
Filters	EMA slow	200	Long-term trend regime.
Filters	VWAP filter (intraday)	true	Long above VWAP / short below VWAP.
Filters	ADX enabled	false	If enabled, requires ADX >= threshold.
Filters	ADX length	14	Directional movement length.
Filters	ADX smoothing	14	Smoothing for ADX line.
Filters	ADX minimum	20	Trend strength threshold.
Benchmark	Relative strength vs SPY	false	Prefers outperformers vs SPY.
Benchmark	RS SMA length	50	Baseline smoothing for RS comparison.
Benchmark	Market regime vs SPY EMA	false	Aligns trades with SPY regime.
Risk	ATR length	14	Used for SuggestedStopPrice (display only).
Risk	ATR multiplier	2.0	Higher = wider stop suggestion.
Visuals	Show levels/EMAs/VWAP	true	On-chart plots for visual confirmation.
Visuals	Show markers	true	Shows BO/BD labels at signals.

6. Outputs and What to Expect

6.1 Screener columns (first 10 plots)

The script's first 10 plots are designed to be used as columns/filters in TradingView's Pine Screener. Common filters are: LongSignal = 1 or ShortSignal = 1.

Plot / Column	Meaning
LongSignal (1/0)	1 on a qualified bullish breakout, else 0.
ShortSignal (1/0)	1 on a qualified bearish breakout, else 0.
ActiveLevelLong	Breakout threshold for long (Donchian or ORB/daily level depending on mode).
ActiveLevelShort	Breakout threshold for short.
VolRatio_or_RelVol	Daily: volume ratio vs SMA. Intraday: relative volume vs SMA.
ADX	Trend strength metric (from DMI). Used if ADX filter enabled.
VWAPDev (intraday)	Close/VWAP - 1 (positive above VWAP). NA in daily mode.
RSRatio vs SPY	Relative strength ratio vs SPY, normalized by its moving average.
SuggestedStopPrice	ATR-based stop suggestion (display only).
ModeCode	1 = Daily logic; 2 = Intraday logic.

6.2 Alerts

The script defines two alert conditions: **Breakout Long** and **Breakout Short**. When used as a watchlist alert, TradingView evaluates the same condition for each symbol in your watchlist and can notify you when any symbol triggers.

6.3 Chart visuals (optional)

If enabled, the script plots breakout levels, EMAs, and session VWAP (intraday) and marks signal bars with small BO/BD labels. These visuals are helpful for manual confirmation but are not required for scanning.

Expected behavior

- Daily mode often triggers near multi-week highs/lows; intraday mode triggers earlier but may produce more false breakouts.
- Tight filters (Trend + ADX + RS + Market regime) can reduce signals significantly.
- Breakouts can fail. Use stops and position sizing. Consider checking upcoming earnings/news before trading a signal.

7. Recommended Workflows

7.1 Best practice: scan a watchlist

This script is best used as a scanner to shortlist candidates, then reviewed manually. A typical flow:

- 1) Run Pine Screener on your watchlist (e.g., S&P; 500).
- 2) Filter for LongSignal = 1 (or ShortSignal = 1).
- 3) Sort by VolRatio/RelVol, RSRatio, or VWAPDev (intraday) to prioritize strength.
- 4) Open top candidates on charts and confirm structure, liquidity, and nearby risk events.

7.2 When to run

- **Daily scan:** near or after the close on 1D. Signals confirm at bar close.
- **Intraday scan:** during the regular session on 5m or 15m, especially after the opening range completes.
- **Alerts:** keep a watchlist alert running to be notified when signals occur.

7.3 Single-symbol use

On a single stock chart, the script functions as a visual breakout detector with optional filters and suggested stop levels. However, its strongest advantage is scanning many symbols quickly.

8. Tuning Guide and Troubleshooting

8.1 Tuning presets

Conservative: donLen 55, UseClose ON, Volume ON (≥ 1.5), Trend ON, ADX ON (≥ 20), RS ON, Market ON.

Balanced: donLen 20, UseClose ON, Volume ON (≥ 1.5), Trend ON, ADX OFF, RS optional, Market optional.

Aggressive: donLen 20, UseClose OFF (High/Low), Volume multiplier 1.2, Trend ON or OFF, ADX OFF.

8.2 Common pitfalls

- **No signals:** loosen filters (ADX/RS/Market) or reduce donLen/volume multiplier.
- **Too many signals:** require UseClose, increase donLen, increase volume multiplier, enable Trend and/or ADX.
- **Intraday ORB not forming:** ensure you are on an intraday timeframe and your session window matches the exchange.
- **Compiler error 'ta.adx' not found:** use ADX from ta.dmi() instead.
- **Compiler error 'end of line without line continuation':** follow Pine line-wrapping rules (continuation lines must not start with a multiple of 4 spaces).

References

- TradingView Help Center - TradingView Pine Screener: key features and requirements
- TradingView Help Center - Watchlist alerts: your trading edge
- TradingView Help Center - Introduction to TradingView alerts
- TradingView Pine Script Docs - Line wrapping
- TradingView Pine Script Docs - Writing limitations